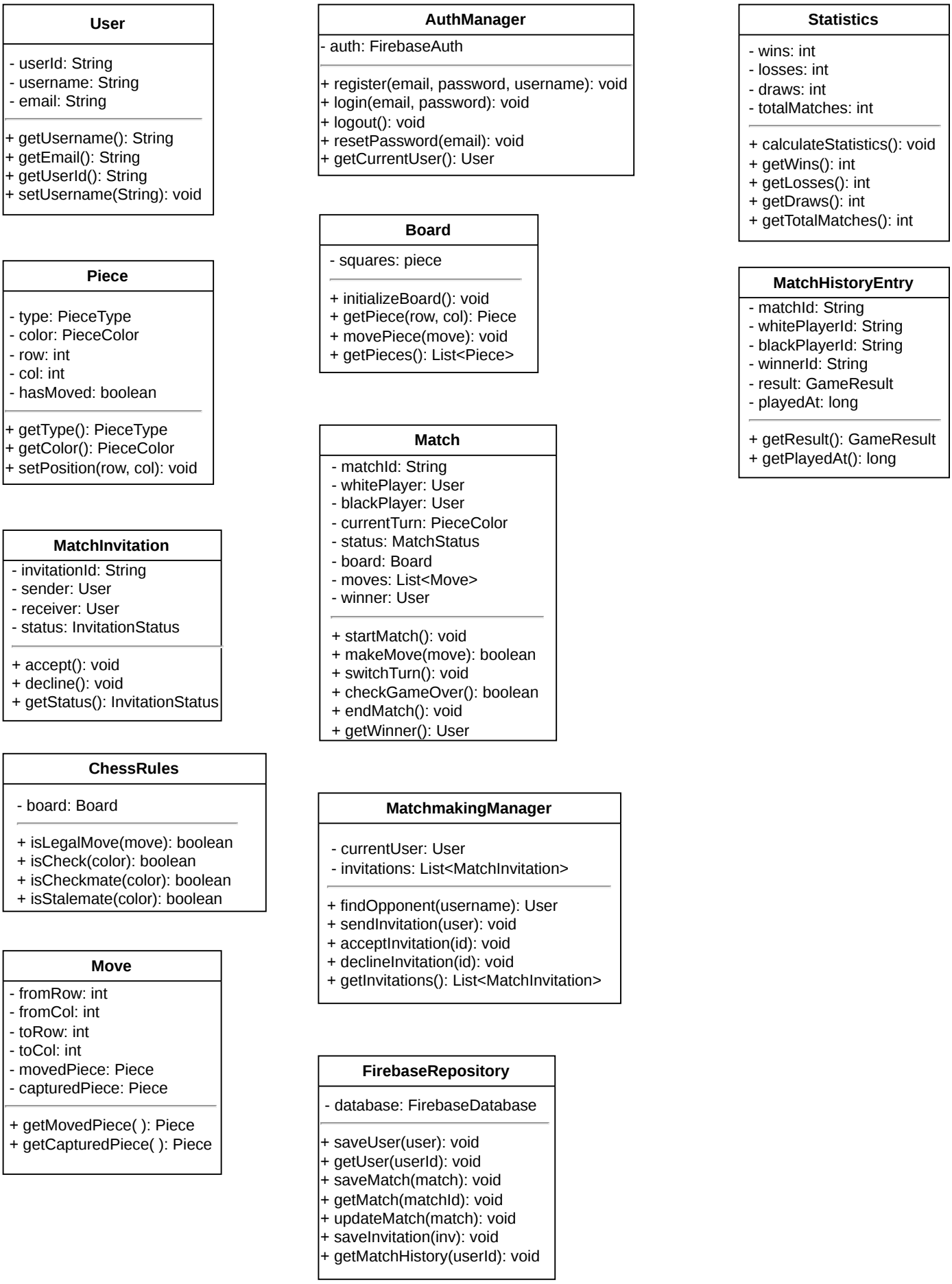
User
- userId: String
- username: String
- email: String
+ getUsername(): String
+ getEmail(): String
+ getUserId(): String
+ setUsername(String): void
AuthManager
- auth: FirebaseAuth
+ register(email, password, username): void
+ login(email, password): void
+ logout(): void
+ resetPassword(email): void
+ getCurrentUser(): User
Statistics
- wins: int
- losses: int
- draws: int
- totalMatches: int
+ calculateStatistics(): void
+ getWins(): int
+ getLosses(): int
+ getDraws(): int
+ getTotalMatches(): int
Board
- squares: piece
+ initializeBoard(): void
+ getPiece(row, col): Piece
+ movePiece(move): void
+ getPieces(): List<Piece>
MatchHistoryEntry
- matchId: String
- whitePlayerId: String
- blackPlayerId: String
- winnerId: String
- result: GameResult
- playedAt: long
+ getResult(): GameResult
+ getPlayedAt(): long
Piece
- type: PieceType
- color: PieceColor
- row: int
- col: int
- hasMoved: boolean
+ getType(): PieceType
+ getColor(): PieceColor
+ setPosition(row, col): void
Match
- matchId: String
- whitePlayer: User
- blackPlayer: User
- currentTurn: PieceColor
- status: MatchStatus
- board: Board
- moves: List<Move>
- winner: User
+ startMatch(): void
+ makeMove(move): boolean
+ switchTurn(): void
+ checkGameOver(): boolean
+ endMatch(): void
+ getWinner(): User
MatchInvitation
- invitationId: String
- sender: User
- receiver: User
- status: InvitationStatus
+ accept(): void
+ decline(): void
+ getStatus(): InvitationStatus
ChessRules
- board: Board
+ isLegalMove(move): boolean
+ isCheck(color): boolean
+ isCheckmate(color): boolean
+ isStalemate(color): boolean
MatchmakingManager
- currentUser: User
- invitations: List<MatchInvitation>
+ findOpponent(username): User
+ sendInvitation(user): void
+ acceptInvitation(id): void
+ declineInvitation(id): void
+ getInvitations(): List<MatchInvitation>
Move
- fromRow: int
- fromCol: int
- toRow: int
- toCol: int
- movedPiece: Piece
- capturedPiece: Piece
+ getMovedPiece( ): Piece
+ getCapturedPiece( ): Piece
FirebaseRepository
- database: FirebaseDatabase
+ saveUser(user): void
+ getUser(userId): void
+ saveMatch(match): void
+ getMatch(matchId): void
+ updateMatch(match): void
+ saveInvitation(inv): void
+ getMatchHistory(userId): void